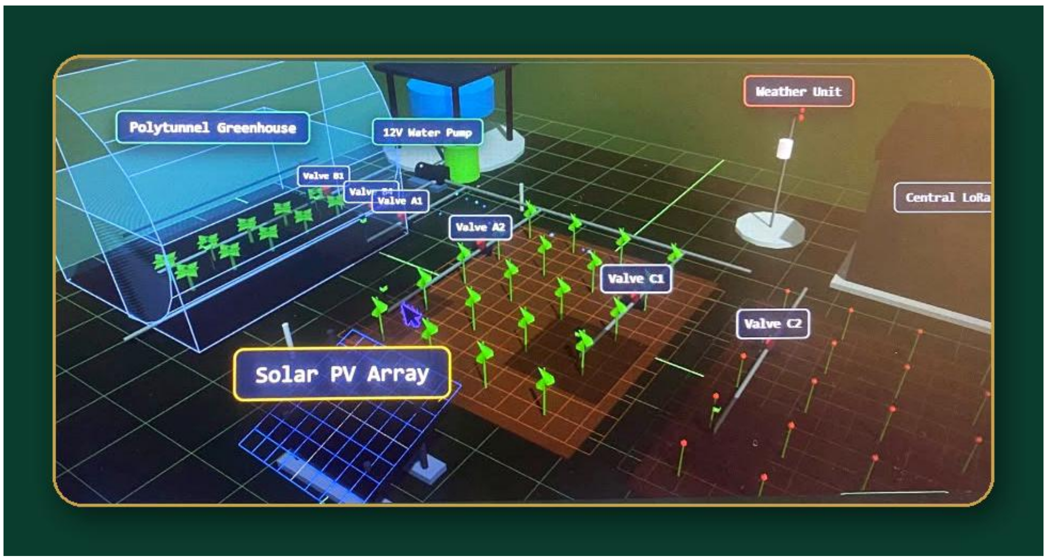
REAL-TIME AGRICULTURAL MONITORING & IRRIGATION MANAGEMENT

AgriSense

Soil Moisture Monitoring and Smart Irrigation Management System



Live 3D Digital Twin — polytunnel greenhouse, irrigation valves, solar PV array and weather unit rendered in real time

"Monitor Smarter. Irrigate Efficiently. Grow Sustainably."

React · Firebase · Firestore · Three.js · Recharts · Tailwind CSS

PROJECT PRESENTATION REPORT

Deployment: Firebase Hosting

Production URL: agrisense-38d3c.web.app

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Executive Summary

AgriSense is a smart agriculture platform designed to improve how farms monitor soil moisture, environmental conditions, irrigation infrastructure, water consumption, and operational events. The system brings monitoring, visualization, irrigation management, analytics, notifications, and farm simulation into one centralized web application.

The current implementation uses React for the web interface, Firebase Authentication for identity management, Cloud Firestore for cloud data and synchronization, Three.js/React Three Fiber for 3D farm visualization, Recharts for analytics, Tailwind CSS for responsive styling, Vite for the development/build process, and Firebase Hosting for deployment.

The platform currently supports telemetry-oriented monitoring, sensor-node management, rule-based irrigation simulation and control, water economics, notifications, authentication, responsive user interfaces, and a 3D digital twin. The next development stage should prioritize physical IoT sensor integration, crop-specific irrigation logic, weather-aware recommendations, safety controls, historical analytics, and machine-learning-assisted irrigation decisions.

Core Value Proposition

> ***AgriSense changes irrigation management from periodic manual observation toward continuous, data-informed monitoring and decision-making.***

Key System Capabilities

- Real-time monitoring of soil moisture, temperature, humidity, ambient light, sensor status, and battery level.
- Interactive 3D digital twin for spatial understanding of farms, zones, sensors, and conditions.
- Sensor node registry for identifying device health and connection status.
- Manual and automated irrigation management for pumps, tanks, valves, and irrigation zones.
- Water consumption, cost, billing, and efficiency monitoring.
- Centralized alerts for low moisture, low tank level, dry-running pumps, offline sensors, low batteries, and irrigation events.
- Firebase-based authentication, Firestore data management, and real-time synchronization.
- Responsive interfaces for desktop, tablet, and smartphone use.

1. Introduction

Agricultural productivity depends strongly on timely and appropriate water management. Traditional irrigation management often depends on physical inspection, fixed schedules, manual records, and separate monitoring of pumps, tanks, valves, and growing areas. These approaches can make it difficult to know the condition of a farm continuously and to respond quickly when water or equipment problems occur.

AgriSense was developed as a centralized digital platform for soil moisture monitoring and smart irrigation management. It combines environmental telemetry, farm visualization, irrigation control, water-use analysis, notifications, authentication, and cloud-based data management in one system.

2. Background and Problem Statement

Farm managers need reliable information about where and when irrigation is required. Without a centralized monitoring system, operators may need to physically inspect multiple areas, estimate soil moisture, check pumps and tanks independently, operate valves manually, and keep separate records of water use.

Problem Statement

There is a need for a centralized system that can continuously monitor soil moisture and environmental conditions, provide visibility of farm infrastructure, support irrigation decisions, track water usage, and notify operators when important events occur. AgriSense addresses this need through an integrated web-based monitoring and management platform.

Problems Addressed

- Limited real-time visibility of farm conditions.
- Manual and delayed soil-moisture assessment.
- Inefficient or unnecessary irrigation.
- Difficulty monitoring pumps, tanks, valves, and sensors.
- Delayed detection of equipment faults.
- Limited historical information for irrigation decisions.
- Difficulty tracking the economic impact of water consumption.

3. Aim and Objectives

3.1 Aim

To develop a real-time agricultural monitoring and irrigation management platform that supports data-driven farm decisions, efficient water use, centralized infrastructure monitoring, and timely response to agricultural events.

3.2 Specific Objectives

1. Monitor soil moisture in real time.

2. Monitor temperature, humidity, and ambient light.
3. Provide centralized monitoring of greenhouse and outdoor farm zones.
4. Track sensor nodes, battery levels, connection status, and operational state.
5. Visualize farm zones and telemetry locations through an interactive 3D digital twin.
6. Provide manual irrigation controls for pumps and valves.
7. Support rule-based automated irrigation based on soil moisture.
8. Track water consumption, irrigation cost, and efficiency indicators.
9. Generate notifications for important farm and equipment events.
10. Provide secure user authentication and controlled access.
11. Synchronize relevant data across active clients in real time.
12. Provide responsive interfaces for desktop, tablet, and mobile devices.

4. Proposed Solution

AgriSense provides a single web application that combines Monitoring, Visualization, Automation, Analytics, and Notifications. Instead of separating soil monitoring, irrigation management, equipment monitoring, and water analysis into different tools, the platform presents them through a unified interface.

Component	Purpose	Value
Telemetry	Collect and display environmental readings	Continuous farm visibility
Irrigation	Manage pumps, valves, tanks and zones	Faster and more controlled watering
Digital Twin	Represent farm areas and sensors spatially	Easier interpretation of farm conditions
Analytics	Track water and environmental trends	Better operational decisions
Notifications	Report important events	Faster response to problems
Authentication	Control user access	Improved security

5. System Scope

5.1 Current Scope

- Web-based farm monitoring.
- Telemetry visualization.
- Sensor-node registry.
- Rule-based irrigation simulation and control.

- 3D digital twin.
- Water economics dashboard.
- Notifications and alerts.
- Firebase authentication and Firestore storage.
- Responsive user interface.
- Cloud deployment.

5.2 Recommended Extended Scope

- Physical IoT sensor integration.
- Physical pump and solenoid-valve control.
- Weather forecast integration.
- Crop-specific moisture requirements.
- Flow-meter-based water accounting.
- Offline-first field operation.
- Machine-learning irrigation recommendations.
- SMS / push notifications.
- Advanced farm reports.

6. System Requirements

6.1 Functional Requirements

13. The system shall authenticate registered users.
14. The system shall display current telemetry for configured farm zones.
15. The system shall register and monitor sensor nodes.
16. The system shall show battery and connection status.
17. The system shall allow authorized users to control pumps and valves.
18. The system shall evaluate irrigation rules.
19. The system shall prevent irrigation when required safety conditions are not satisfied.
20. The system shall record irrigation events and water usage.
21. The system shall generate alerts for configured events.
22. The system shall display historical trends and key performance indicators.
23. The system shall provide responsive interfaces across supported screen sizes.

6.2 Non-Functional Requirements

- **Security —** authentication, authorization, and least-privilege access.
- **Reliability —** safe handling of missing, delayed, or invalid sensor readings.
- **Performance —** efficient dashboard rendering and database queries.

- **Scalability —** support for additional farms, zones, and sensor nodes.
- **Usability —** clear status indicators and simple farm operations.
- **Maintainability —** modular React components and centralized business logic.
- **Availability —** cloud deployment with reliable synchronization.
- **Safety —** irrigation controls must include fail-safe conditions.

7. System Architecture

The application is structured into presentation, application/business-logic, visualization, and Firebase service layers.

Layer	Technology / Component	Responsibility
Presentation	React 18, Tailwind CSS, CSS Grid, Flexbox	User interface and navigation
Application	AuthContext, SimulationContext, business logic	State, rules, and application behavior
Visualization	Three.js, React Three Fiber, Drei, Recharts	3D visualization and analytics
Backend Services	Firebase Authentication	Identity and access
Data	Cloud Firestore	Farm, telemetry, state, and notification data
Hosting	Firebase Hosting	Production deployment

The architecture can be extended to include an IoT ingestion layer between physical sensor nodes and Firestore. For a production agricultural deployment, this layer should also validate telemetry, manage device identity, handle intermittent connectivity, and protect control commands.

8. Major Functional Modules

8.1 Dashboard

Provides a centralized view of current farm conditions, key indicators, equipment state, alerts, and irrigation activity.

8.2 Telemetry Monitoring

Displays soil moisture, temperature, humidity, ambient light, sensor status, and battery information.

8.3 Sensor Node Registry

Maintains device identity, location, battery, connectivity, operational state, and associated telemetry.

8.4 Irrigation Management

Supports pump, tank, valve, and zone monitoring, including manual controls and automated rules.

8.5 Water Economics

Tracks water consumption, cost, billing information, trends, and efficiency indicators.

8.6 Notifications

Centralizes alerts such as low moisture, low tank level, dry-running pump, sensor disconnection, and low battery.

8.7 Authentication

Provides secure login and user identity management through Firebase Authentication.

9. 3D Digital Twin

The 3D Digital Twin is a distinctive component of AgriSense. It provides a virtual representation of the farm using Three.js, React Three Fiber, and Drei. The existing design includes terrain, zones, telemetry points, and sensor locations. A rendered snapshot of the live twin is presented later in this report.

Recommended Enhancement

- Use color-coded zones based on soil moisture status.
- Allow users to click a zone and open its live telemetry.
- Show pump and valve state directly on the model.
- Animate active irrigation zones.
- Display alerts spatially at the affected sensor or equipment.
- Provide a simple 2D fallback for low-powered mobile devices.

10. Telemetry and Sensor Management

The telemetry layer is intended to provide continuous environmental information. Current parameters include soil moisture, temperature, humidity, ambient light, sensor status, and battery level.

Recommended Physical IoT Architecture

Device	Purpose
ESP32	Low-cost field controller and communication device
Capacitive soil-moisture sensor	Measure soil moisture with improved durability over basic resistive probes
Temperature / humidity sensor	Measure environmental conditions
Water-level sensor	Determine available water in tanks
Flow meter	Measure actual water consumption
Relay / driver	Control pump or valve circuitry safely
Solenoid valve	Control individual irrigation zones

The physical device layer should send timestamped readings and receive authorized control commands. The system should validate readings before using them for automatic irrigation.

11. Smart Irrigation Management

The current rule-based model starts irrigation when soil moisture falls below a configured threshold, checks water availability, and stops irrigation when the target moisture level is reached.

Improved Decision Model

- Check whether the sensor reading is recent and valid.
- Compare moisture with the crop and growth-stage threshold.
- Check available tank water.
- Check whether rainfall is expected.
- Check whether the zone is permitted to irrigate at the current time.
- Check recent irrigation history to avoid repeated short cycles.
- Start the pump and open the correct valve when conditions are satisfied.
- Monitor flow and moisture while irrigation is active.
- Stop irrigation at the target condition or safety limit.
- Record the event and generate an appropriate notification.

Important safety principle: physical irrigation should fail safely. A missing sensor reading, empty tank, abnormal flow, excessive runtime, or communication failure should never cause an uncontrolled pump to remain running.

12. Water Economics and Analytics

The Water Economics module should convert technical telemetry into information that a farm manager can use for financial and operational decisions.

- Daily, weekly, and monthly water consumption.
- Water consumption by zone.
- Pump runtime by zone.
- Estimated irrigation cost.
- Water-use efficiency.
- Comparison with previous periods.
- Identification of abnormal consumption.

- Estimated water and cost savings.

A flow meter should be added to make water-use estimates more reliable. This would allow the system to compare expected and actual water delivery and support leak or abnormal-consumption detection.

13. Notifications and Safety

The current notification concept includes low moisture, low tank level, pump dry-running, sensor disconnection, low battery, irrigation triggered, and irrigation completed.

Recommended Alert Priorities

Priority	Example	Recommended Action
Critical	Pump running with no flow	Automatic shutdown + immediate alert
High	Tank critically low	Prevent new irrigation and alert operator
High	Sensor offline during automation	Disable affected automatic rule
Medium	Low battery	Schedule maintenance
Low	Irrigation completed	Informational notification

14. Authentication and User Roles

Firebase Authentication provides the foundation for user identity. The system should be extended with role-based access so that users can only perform actions appropriate to their responsibilities.

Role	Typical Permissions
Administrator	Manage users, farms, zones, devices, rules, and system configuration
Farm Manager	Monitor farms, control irrigation, configure zones, view reports
Farm Worker	View assigned zones, receive alerts, perform permitted operational actions
Viewer	Read-only monitoring and reports

An audit log should record important actions such as pump activation, valve changes, threshold updates, user-management changes, and manual overrides.

System Snapshot — The 3D Digital Twin in Action

A rendered view of the live AgriSense 3D Twin Simulator (see Section 9), showing the polytunnel greenhouse, irrigation valve network (A1/A2, B1/B2, C1/C2), the 12V water pump, solar PV array, and weather unit.

The AgriSense 3D Twin — a live spatial model of farm zones, equipment, and sensor nodes.

15. Database Architecture

Cloud Firestore is used to store and synchronize farm data, telemetry, sensor nodes, simulation state, irrigation state, notifications, and user-related information.

Recommended Logical Structure

Collection	Example Data
users	Profile, role, assigned farms
farms	Farm details, location, configuration
zones	Crop, soil type, thresholds, valve, sensor references
sensorNodes	Device ID, battery, status, last seen
telemetry	Timestamped moisture, temperature, humidity, light
irrigationEvents	Start, stop, duration, zone, reason
equipment	Pumps, tanks, valves, flow meters
notifications	Type, priority, status, timestamp
auditLogs	User, action, target, timestamp

16. Sensor-to-Dashboard Data Flow

The recommended end-to-end flow is: Farm environment → physical sensor → field controller → secure network connection → data ingestion / validation → Firestore → AgriSense web application → visualization, analytics, alerts, and decision logic. Authorized irrigation commands should follow the reverse path from the application to the controller and physical equipment.

17. Technology Stack

Technology	Role in AgriSense
React 18	Frontend application
Tailwind CSS	Responsive styling
Firebase Authentication	User authentication
Cloud Firestore	Cloud database and synchronization
Three.js / React Three Fiber / Drei	3D digital twin
Recharts	Charts and analytics
Vite	Development and build tooling
Firebase Hosting	Deployment

For physical deployment, an IoT communication layer such as MQTT or an HTTPS-based device API can be introduced. The final choice should depend on connectivity, security, device count, and operational requirements.

18. User Interface and UX

The project already emphasizes mobile-first design, responsive grids, off-canvas navigation, touch-friendly controls, and fluid typography.

Recommended UX Improvements

- Place the most important farm status information above the fold.
- Use consistent status colors and labels across all modules.
- Show 'last updated' time for every live telemetry card.
- Clearly distinguish simulated values from physical sensor values.
- Use confirmation dialogs for high-impact manual controls.
- Display the current farm, zone, and crop context throughout the dashboard.
- Provide simple explanations for irrigation recommendations.
- Add a compact mobile control panel for emergency shutdown and critical alerts.

19. Security and Reliability

- Use Firestore Security Rules based on authenticated identity and role.
- Restrict users to farms and zones they are authorized to access.
- Validate all device telemetry before storing or acting on it.
- Never trust a client-side permission check as the only security control.
- Protect device credentials and control endpoints.
- Log manual irrigation overrides and configuration changes.
- Use server timestamps where appropriate.
- Implement rate limits or command validation for physical control actions.
- Provide safe defaults when connectivity or sensors fail.

20. Testing and Evaluation

The improved project should include evidence-based testing rather than only feature descriptions.

Test Area	Example Test	Expected Result
Authentication	Login with valid and invalid credentials	Valid user enters system; invalid login rejected
Telemetry	Submit new sensor reading	Dashboard updates with correct timestamp
Offline node	Disconnect a sensor	Node becomes offline and alert is generated
Low moisture	Set moisture below threshold	Irrigation rule is evaluated
Low tank	Set tank below minimum	Irrigation is prevented and alert generated
No-flow	Run pump without detected flow	Pump is stopped and critical alert generated
Mobile UX	Test common phone widths	Controls remain usable and readable
Security	Access another user's farm	Access is denied

Where physical hardware is available, record measured response times, sensor accuracy, synchronization latency, irrigation runtime, water consumption, and system uptime. These results can be included in the final evaluation chapter.

21. Deployment and Maintenance

The current project is deployed through Firebase Hosting. A production workflow should include development, testing/staging, and production environments.

- Use version control for all application and configuration code.
- Maintain separate Firebase projects or controlled environments for development and production.
- Back up important data and configurations.
- Monitor application errors and device failures.
- Document sensor calibration and hardware replacement procedures.
- Keep dependencies updated and review security advisories.
- Maintain a device registry and installation record.

22. Current Limitations

Based on the supplied report, the project currently emphasizes web-based monitoring, simulation, and cloud management. The report does not provide evidence that physical sensors, physical pump/valve control, weather services, or a trained machine-learning model are already integrated.

- Physical IoT integration is identified as future work.
- AI-based irrigation prediction is identified as future work.
- Weather integration is identified as future work.
- Advanced reporting is identified as future work.
- The report does not present quantitative field-test results.
- The report does not document measured irrigation savings from a real farm deployment.

23. Recommended Improvements

#	Improvement	Reason
1	Physical IoT sensor integration	Moves the system from simulation toward real-world deployment
2	Physical pump and valve integration	Demonstrates end-to-end automated irrigation
3	Crop-specific irrigation rules	Makes recommendations more agriculturally meaningful
4	Weather integration	Prevents irrigation when rainfall is likely
5	Flow-meter integration	Improves water accounting and leak detection
6	Safety / fail-safe controls	Protects equipment and crops
7	Historical analytics	Supports evidence-based decisions
8	AI irrigation recommendation	Adds predictive intelligence after reliable data collection

#	Improvement	Reason
9	Role-based access and audit logs	Strengthens security and accountability
10	Offline field operation	Improves suitability for unreliable connectivity

24. Future Development Roadmap

Phase 1 — Hardware Prototype

- Connect ESP32 to soil-moisture and environmental sensors.
- Send timestamped telemetry to the cloud.
- Display real readings in the dashboard.

Phase 2 — Automated Irrigation

- Connect water-level and flow sensors.
- Control a pump and one or more valves safely.
- Implement emergency shutdown rules.

Phase 3 — Decision Intelligence

- Add crop profiles and growth stages.
- Integrate weather forecasts.
- Use historical telemetry to recommend irrigation.

Phase 4 — AI and Prediction

- Train a model using local farm data.
- Predict soil moisture and irrigation requirements.
- Detect abnormal sensor / equipment behavior.
- Evaluate model accuracy against real irrigation outcomes.

Phase 5 — Scale and Commercialization

- Multi-farm management.
- Subscription or service-based pricing.
- SMS / push notification services.
- Installation and maintenance support.
- Agronomist or extension-worker integration.

25. Expected Impact

- Improved visibility of soil and environmental conditions.
- More timely irrigation decisions.
- Reduced unnecessary water use.

- Faster detection of equipment faults.
- Better tracking of water and irrigation costs.
- Improved accountability through event and audit records.
- A foundation for data-driven and predictive agricultural management.

The actual magnitude of these benefits should be established through controlled field trials comparing AgriSense-assisted irrigation with an appropriate baseline method.

26. Project Innovation

The main innovation of AgriSense is not a single technology but the integration of several capabilities into one agricultural management environment.

- Real-time telemetry.
- Interactive 3D farm visualization.
- Rule-based smart irrigation.
- Water economics.
- Cloud synchronization.
- Event-driven notifications.
- Responsive farm management interface.
- A clear pathway toward physical IoT and AI-assisted irrigation.

27. Conclusion

AgriSense is a strong foundation for a smart agriculture and irrigation-management platform. The current system demonstrates how modern web technologies can centralize telemetry, farm visualization, irrigation management, water analytics, notifications, authentication, and cloud data synchronization.

The most important next step is to connect the software to physical agricultural equipment and reliable field data. Physical sensors, flow measurement, pump and valve control, crop-specific rules, weather-aware decision-making, safety mechanisms, and machine-learning recommendations would transform AgriSense from a feature-rich software prototype into a stronger real-world smart irrigation solution.

Final Project Direction

Simulation → Physical IoT → Safe Automation → Weather-Aware Decisions → AI Prediction → Measured Farm Impact

AgriSense — Monitor Smarter. Irrigate Efficiently. Grow Sustainably.

28. Presentation Summary

An 18-slide condensed version of this report, suitable for a project defense or stakeholder pitch.

#	Topic	Content
1	Title	AgriSense: Soil Moisture Monitoring and Smart Irrigation Management System.
2	Problem	Manual inspection, inefficient irrigation, water wastage, delayed fault detection, and limited real-time information.
3	Proposed Solution	A centralized platform for monitoring, visualization, irrigation, analytics, and notifications.
4	Objectives	Monitor moisture and environmental conditions; manage irrigation; track sensors; analyze water; provide alerts.
5	Main Features	Telemetry, 3D Digital Twin, Node Registry, Smart Irrigation, Water Economics, Notifications, Authentication.
6	Architecture	React application connected to Firebase Authentication and Firestore, with visualization and business logic.
7	Digital Twin	Spatial representation of farm zones, sensor locations, telemetry, and equipment.
8	Telemetry	Soil moisture, temperature, humidity, light, battery, and connection status.
9	Irrigation	Moisture threshold, water availability, pump/valve operation, target moisture, and safety rules.
10	Water Economics	Consumption, cost, efficiency, trends, and future flow-meter integration.
11	Notifications	Low moisture, low tank, no-flow, sensor offline, low battery, and irrigation events.
12	Security	Authentication, role-based access, audit logs, and Firestore security rules.
13	IoT Roadmap	ESP32, soil sensors, water-level sensor, flow meter, pump and solenoid valves.
14	AI Roadmap	Crop-specific recommendations, weather-aware irrigation, prediction, and anomaly detection.
15	Testing	Functional, security, performance, mobile, sensor, irrigation, and safety tests.
16	Deployment	Vite build and Firebase Hosting production deployment.

#	Topic	Content
17	Impact	Efficient water management, faster response, better monitoring, and data-driven decisions.
18	Conclusion	AgriSense provides a foundation for real-time, safe, intelligent, and scalable irrigation management.